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**ТЕМАТИЧЕСКАЯ И ЭМОЦИОНАЛЬНАЯ ДИНАМИКА В
ТЕКСТАХ ФРАНЦУЗСКИХ ПЕСЕН: ВЫЧИСЛИТЕЛЬНЫЙ
NLP-АНАЛИЗ С 1925 ПО 2025 ГОД
МАГИСТЕРСКАЯ ДИССЕРТАЦИЯ**

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**THEMATIC AND EMOTIONAL TRENDS IN FRENCH SONG LYRICS: A
COMPUTATIONAL NLP ANALYSIS FROM 1925 TO 2025**

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ABSTRACT

This thesis traces how themes and emotional tones in French song lyrics shifted between 1925 and 2025. Most computational work on popular music has either stayed with small qualitative samples or focused on English-language material, and French popular song has remained largely outside that conversation. The corpus was built from MusicBrainz metadata, enriched with lyrics from Genius, then cleaned, deduplicated and prepared for NLP analysis. The classification-ready dataset contains 63,652 songs.

Classification is done by zero-shot natural language inference, using two multilingual transformer models to assign eight topic labels and eight mood labels. The outputs are validated through template-sensitivity checks, cross-model agreement, human annotation, confidence calibration, bootstrap uncertainty estimates and genre-adjusted regression. The cautious design is not optional. Song lyrics are noisy and metaphorical, they repeat themselves, and the genre composition of the corpus changes substantially across decades.

The central finding is that French lyrics do not simply become "more negative" over time – the emotional structure changes. Anxiety rises sharply, while tenderness, joy and direct sadness decline. Thematically, the strongest long-term increase is in nostalgia, memory and the passage of time. Love remains important but loses its dominance, partly because rap and contemporary pop take up more space in the corpus from the 2000s onward. The apparent rise of protest and social critique is largely explained by that same shift toward rap, rather than by any uniform change across genres. The broader point is methodological: computational NLP can surface large-scale cultural patterns in French song lyrics, but only when uncertainty, bias and the limits of the method are kept visible rather than smoothed over.

Keywords: French song lyrics; chanson; NLP; cultural analytics; topic classification; emotion analysis; zero-shot learning; MusicBrainz; Genius; diachronic corpus.

Аннотация

В данной диссертации прослеживается эволюция тематики и эмоциональной окраски текстов французских песен в период с 1925 по 2025 год. Большинство исследований в области популярной музыки либо ограничивались небольшими качественными выборками, либо фокусировались на англоязычном материале, а французская популярная песня в основном оставалась за пределами этой дискуссии. Корпус был построен на основе метаданных MusicBrainz, обогащен текстами песен из Genius, а затем очищен, дедулицирован и подготовлен для анализа с помощью методов обработки естественного языка (NLP). Набор данных, готовый к классификации, содержит 63,652 песни.

Классификация осуществляется с помощью инференции естественного языка методом zero-shot, с использованием двух многоязычных трансформаторных моделей для присвоения восьми тематических меток и восьми меток настроения. Результаты проверяются с помощью проверок чувствительности к шаблонам, согласованности между моделями, аннотации людьми, калибровки доверия, оценок неопределенности методом бутстрапа и регрессии с поправкой на жанр. Такой осторожный подход не является факультативным. Тексты песен содержат много шума и метафор, они повторяются, а жанровый состав корпуса существенно меняется на протяжении десятилетий.

Основной вывод заключается в том, что тексты французских песен со временем не просто становятся «более негативными» – меняется их эмоциональная структура. Уровень тревоги резко возрастает, в то время как нежность, радость и прямая печаль снижаются. С тематической точки зрения наиболее заметный долгосрочный рост наблюдается в таких темах, как ностальгия, воспоминания и течение времени. Любовь по-прежнему остается важной темой, но теряет свое доминирующее положение, отчасти потому, что с 2000-х годов в корпусе все больше места занимают рэп и современная поп-музыка. Очевидный рост протеста и социальной критики в значительной степени объясняется тем же сдвигом в сторону рэпа, а не каким-либо единообразным изменением во всех жанрах. Более общий вывод носит методологический характер: вычислительный НЛП может выявить крупномасштабные культурные паттерны в текстах французских песен, но только в том случае, если неопределенность, предвзятость и ограничения метода остаются видимыми, а не сглаживаются.

Ключевые слова: тексты французских песен; шансон; NLP; культурная аналитика; классификация тем; анализ эмоций; обучение без обучающих данных; MusicBrainz; Genius; диахронический корпус.

Chapter 1. INTRODUCTION

1.1. Relevance and context

Song lyrics sit in an unusual position: they are poetry, they are everyday speech, and they are social documents. As a hybrid genre situated between literature and popular culture, songs both articulate individual experience and remain deeply embedded in the zeitgeist. In France, the tradition of the *chanson* has long carried the language people actually used about love, work, politics and loss. That makes lyrics a useful empirical window into how a culture talks to itself across time. From a linguistic perspective, they offer access to vernacular usage, metaphorical patterns, and evaluative language that are often underrepresented in canonical literary corpora, while from a cultural perspective they serve as markers of collective identity and memory.

Most existing work on French song operates at a small scale: a single artist, a single decade, a single genre. Quantitative studies on English-language pop have been more ambitious, and they converge on a few clear findings. Dodds and Danforth [2] reported “a systematic decrease in the happiness of song lyrics over the period 1960–2007” across 232,574 songs. DeWall et al. [4] later showed that U.S. hit lyrics had become “more self-focused, more angry, and less positive” between 1980 and 2007. Parada-Cabaleiro et al. [8] documented that “song lyrics have become significantly more repetitive over the last five decades”. None of these results have been replicated systematically for French.

By combining diachronic corpus construction with computational analysis and historically informed interpretation, this research responds to both a substantive gap in the study of French song lyrics and a broader demand for methodologically rigorous, interdisciplinary approaches to cultural data. Using a century of French-language song lyrics (1925–2025), it asks how the topics and emotional registers of those lyrics have shifted, and how confident we can be in any answer. The full code and notebooks are open: <https://github.com/daniilmv/hsemasters> [31].

1.2. Object, subject and aim

The **object** of the research is French song lyrics, treated as a linguistic and cultural phenomenon: texts that combine poetry, ordinary speech and culturally situated forms of expression. They function simultaneously as artistic productions and as social documents, reflecting norms, values, emotions, and representations circulating within French society at specific historical moments. As such, French song lyrics provide a relevant empirical basis for studying language use, thematic preoccupations, and

emotional expression across time, while also raising methodological questions related to genre, performativity, and cultural context.

The **subject** is the evolution of thematic and emotional patterns in those lyrics from 1925 to 2025, examined through corpus-based NLP. The focus lies on identifying recurrent topics, dominant emotional orientations, and their temporal dynamics within a large diachronic corpus. By operationalizing themes and emotions as measurable textual patterns, the study aims to analyze how collective concerns and affective regimes change over the course of a century, while accounting for historical ruptures, continuities, and shifts in cultural production.

The **aim** is to identify and analyse long-term patterns in topics and moods in French song lyrics across the period, combining quantitative modelling with historically informed interpretation.

To achieve this aim, the research pursues the following **objectives**: build a diachronic corpus from open digital sources; normalise the lyrics; classify them by topic and mood using methods appropriate for poetic French; aggregate by decade; test for trends; and contextualise the results without overclaiming. Quantitative results might be interpreted against major social, political, and cultural developments, allowing for the examination of links between textual trends and broader historical processes. Finally, we strive to visualize and interpret thematic and emotional trends in relation to social change.

1.3. Research questions and hypothesis

Two research questions guide the work:

***Q1.** How have the dominant themes in French song lyrics evolved across the twentieth and twenty-first centuries?*

***Q2.** How do emotional tones in French song lyrics correspond to major historical and socio-political transformations?*

These questions aim to capture both the semantic content of song lyrics and their affective dimension, treating themes and emotions as complementary indicators of cultural change. The diachronic perspective allows for the identification of periods of stability, transition, and rupture in lyrical content, while also enabling comparison across distinct historical contexts.

The working **hypothesis** is that periods of social and political turmoil are associated with more negative emotional tones and more reflective or introspective lyrical content. This hypothesis is tested empirically through computational analysis and statistical validation, with the objective of determining

whether observed patterns correspond systematically to moments of crisis or transformation rather than to random variation in the data. As shown in Chapter 3, the evidence supports parts of this and forces a reformulation of the rest: the corpus does not become uniformly more negative, but the form of negative affect shifts.

1.4. Methods overview

The corpus is built in four stages. First, MusicBrainz [22] provides a structured metadata backbone covering tracks, artists, releases, countries, languages and genres. Second, a composite French-likelihood score filters that backbone down to a high-recall candidate set. Third, Genius (queried through the LyricsGenius client [23]) supplies the lyric texts via title-and-artist matching. Fourth, the resulting pool is cleaned, deduplicated and prepared for modelling.

For classification, the actual pipeline does not use the lexicon-plus-LDA approach common in earlier work. With no labelled training set of French lyrics available, the study uses zero-shot natural-language-inference classification [24] with two multilingual models: mDeBERTa-v3-base-mnli-xnli as the primary classifier (M1) and XLM-RoBERTa-large-xnli (M2) for cross-validation. Each lyric is treated as a premise; each topic or mood label is inserted into a French-language hypothesis template. Eight topic labels and eight mood labels are tracked across decades.

Outputs are validated through five independent diagnostics: template sensitivity, cross-model agreement (Cohen’s κ , Spearman ρ , Jaccard), a small 182-song human annotation gold standard, confidence calibration, and genre-adjusted logistic regression. Decade-level shares are then tested for monotonic trend with the Mann–Kendall test [25, 26] and Sen’s slope [28], with Benjamini–Hochberg FDR correction [27]. Artist-clustered bootstrap (1,000 iterations) gives the uncertainty intervals. The implementation uses Python with transformers, pandas, scikit-learn, statsmodels, pymannkendall, matplotlib and seaborn.

1.5. Scientific novelty and practical significance

The **scientific novelty** of this research lies in the development of the first large-scale, data-driven model of the evolution of French song lyrics that systematically combines natural language processing, cultural analytics, and historical interpretation over a century-long period. While previous studies have addressed French song either through qualitative analysis or limited quantitative approaches, none have proposed a diachronic framework of comparable temporal scope grounded in reproducible computational methods. By constructing and analyzing a large corpus spanning from 1925 to 2025, this study moves

beyond artist-centered or period-specific analyses and instead captures structural patterns and long-term dynamics in thematic and emotional expression.

Practically, the corpus and pipeline are reusable. The repository is public [31], and the cleaned dataset and the song-level predictions can be applied to musicology, French linguistics and cultural-analytics teaching. A public-facing companion site (<https://daniilvoloshin.com/chanson>) presents the findings as an interactive narrative for non-specialist readers, as described in Section 3.8. Additionally, the approach has applications in archival research and media studies, where large collections of cultural texts increasingly require computational tools for exploration, classification, and interpretation. More broadly, the research demonstrates how data-driven methods can enrich the analysis of cultural heritage materials while remaining sensitive to historical and social context.

1.6. Related work

English-language work converges on a fairly consistent story: lyrics have become less happy, more self-focused and more lexically repetitive. Napier and Shamir [3] analysed 6,150 Billboard Hot 100 songs (1951–2016) and found that words conveying joy or confidence dropped steadily, while anger, disgust and sadness gained ground. Brand, Acerbi and Mesoudi [5] traced fifty years of English lyric emotion and described the same broad drift. Varnum et al. [6] reported a parallel decline in lexical complexity across six decades of American pop. Foramitti et al. [13] caution that this picture is not flat: “societal crises may temporarily disrupt” the long-term trend, so charts deserve more than a fitted line.

Outside English the picture diverges. Hunke et al. [10] tracked German popular music from 1954 to 2022 and found a similar drop in positive sentiment, alongside a shift away from dreams-and-longing toward society-and-status. Jo and Kim [7] reported the opposite for K-pop (1990–2019), describing a “significantly more positive emotional trajectory” than its Western counterparts. Wang and Wong [11] paired an LLM with a Chinese emotion lexicon on 264,851 Mandarin pop songs (1967–2023), and a companion paper [12] modelled the resulting series as a damped oscillator with “a sine-wave-like pattern with an estimated period of 34 to 35 years”.

French has been studied less, and mostly outside computation. Hammou’s history of French rap [30] gives the cultural-historical context – rap as a politicised lyrical space built in the banlieues from the late 1980s onward. Computationally, Zurbuchen and Voigt [9] are an obvious anchor: working from 8,222 French rap songs (1991–2024), they show that “Anglicisms and other contact-induced borrowings are pervasive and increasing in French rap”. Beyond that, large-scale diachronic studies of French lyrics are essentially absent.

Method choices in this literature track the same constraints. Early studies used lexicons, LIWC categories [4], the NRC Emotion Lexicon [15], which are transparent but blind to irony, slang and metaphor. Newer work uses transformer-based classifiers, but these have their own failure modes on poetic text. Manovich [29] frames the broader project of cultural analytics as one of complementing – not replacing – close reading with large-scale pattern analysis. The pipeline here is shaped by that constraint: classification is a structured approximation, not a final reading, and the outputs are validated before being interpreted.

Two further methodological points matter for French. First, French songs increasingly mix English, Arabic and *verlan*; a unified classifier has to tolerate code-switching. Second, the genre composition of any French lyric corpus changes drastically across decades – *chanson*, *yéyé*, *variété*, rock, rap – and any temporal trend may track that compositional shift rather than a within-genre change. Both issues are returned to in Chapter 2.

1.7. Thesis overview

The thesis is organized to ensure a logical and progressive development of the research, moving from conceptual framing to empirical analysis and interpretative discussion.

Chapter 1 provides the theoretical and methodological foundations of the research. It conceptualizes song lyrics as linguistic and cultural objects, examines their specific textual and stylistic characteristics, and reviews key approaches in corpus linguistics, computational text analysis, and digital humanities. This chapter also discusses previous research on song lyrics and identifies methodological challenges inherent in the automated analysis of artistic and historically situated texts.

Chapter 2 is devoted to corpus construction and analytical methods. It details the sources of the lyric data, the principles guiding corpus design, and the preprocessing steps applied to ensure linguistic consistency and diachronic comparability. The chapter further explains the natural language processing techniques, models, and tools employed in the study, with particular attention to transparency, reproducibility, and potential biases in the data.

Chapter 3 presents the results of the thematic and emotional analyses. It examines the evolution of dominant themes and emotional patterns in French song lyrics across the twentieth and twenty-first centuries, relying on statistical modeling and visualization to identify long-term trends, periods of stability, and moments of change. The emphasis is placed on empirical findings derived directly from the computational analyses. The chapter also offers primary cultural interpretation and critical discussion of these results. The quantitative patterns identified in the previous chapter are contextualized within

broader historical, social, and cultural developments in France. This chapter evaluates the extent to which the findings support the proposed hypothesis and reflects on the limitations of the methods and data.

The conclusion synthesizes the main results of the study, revisits the research questions and objectives, and highlights the theoretical and methodological contributions of the thesis. It also outlines the study's limitations and suggests directions for future research in the computational analysis of cultural and linguistic data.

Chapter 2. Corpus Construction and Methods

2.1. Pipeline overview

No ready-made corpus of French song lyrics 1925–2025 existed in the public domain. The pipeline built one in four stages, with a deliberate funnel shape: early stages prioritise coverage, later stages prioritise reliability. Stage 1 pulls a broad metadata universe from MusicBrainz. Stage 2 narrows it to likely-French candidates using a composite score. Stage 3 attaches lyrics from Genius. Stage 4 cleans, deduplicates and prepares the result for classification.

The reductions across these stages are large, and they are not accidents. Each filter is documented and reproducible. Table 1 reports the row counts at every stage.

Table 1 – Corpus construction stages and row counts – how the row count changes and why

Stage	Operation	Rows retained	Function
Initial MusicBrainz pool	Track-level metadata extraction	121,327,699	Broad metadata universe
France-related subset	Country-based filtering	6,892,156	France-connected repertoire
Likely French candidates	French-likelihood score ≥ 3	4,186,516	High-recall French registry
Genius-enriched records	Title/artist match through Genius	1,082,070	Enriched metadata pool
Records with lyrics	Successful lyric extraction	81,823	Lyrics-enriched pool
Curated NLP dataset	Standardised analytical schema	66,711	Working corpus
Model-ready corpus	Cleaned, non-empty lyrics	65,230	Input to classification
Classification-ready	Deduplicated, token-filtered	63,652	Final input for topic/mood

The final corpus is not a complete archive of French songs from 1925 to 2025. It is a large, metadata-informed, lyrics-available corpus. Its strengths are size, reproducibility and rich metadata. Its limitations – uneven historical coverage, dependence on what Genius has, and a heavy genre skew toward rap and pop in later decades – are addressed in the sections that follow.

2.2. MusicBrainz registry

MusicBrainz [22] was the metadata backbone. It is an open music encyclopedia with structured tables for artists, recordings, releases, works, countries, languages and genres. It does not provide lyrics – that is not what it was needed for. What it provides is something better for filtering: relational metadata, downloadable as a database dump, that ties recordings to artists, releases, countries, languages and genres through explicit relationships.

The first notebook (*00_musicbrainz_french_artist_registry*) loads the core tables – tracks, media, releases, recordings, artist credits, names, languages, countries, release countries, genres, artist relationships – and joins them into a candidate set with title, artist, year, language, release country, artist country, gender and genre. The initial pool contains 121,327,699 track-level rows; most have no connection to France.

A few caveats about MusicBrainz are worth naming upfront. It is community-maintained, so completeness varies by artist, country and decade. Songs may appear multiple times because of reissues, compilations and alternate recordings. A French release country does not guarantee French lyrics; a French artist country does not mean every song is in French. For these reasons, no single metadata field is treated as definitive.

2.3. French candidate selection

Country filtering alone is not enough. A composite French-likelihood score combines four signals – language metadata, release country, artist country and a title-level cue – and records scoring at least 3 are retained as likely French candidates. The threshold balances recall and precision: lower kept too much noise, higher cut too many real French songs with incomplete metadata. The weights are shown in Table 2.

Table 2 – French-likelihood score components

Signal	Score	Rationale
French language metadata	+3	Direct evidence of French-language content
French release country	+2	Indicates distribution/publication in France
French artist country	+2	Indicates artist-level connection to France
French-like title signal	+1	Captures title-level linguistic cues

Filtering on this score produced 4,186,516 candidate rows – still a registry, not a corpus, but a high-recall starting point for lyric retrieval. The registry preserved the song title, artist, artist credit, year, language, release country, artist country, genre and the score itself, with genres in both a long and an aggregated format.

2.4. Genius enrichment

Genius is the main source of lyric texts. Its songs pages can be queried by title and artist; the lyricsgenius client [23] wraps the API. The enrichment notebook (01_musicbrainz_enrichment) iterates over the candidate registry and queries Genius for each row. Because exact string matching between MusicBrainz and Genius is unreliable – artist credits vary because of collaborations, aliases and spelling differences – matching uses a similarity score with title weighted at 65% and artist at 35%.

Processing used chunked execution, checkpointing, retry logic and a resilient HTTP session to handle rate limits and intermittent failures – necessary because the input contained millions of candidates and a single uninterrupted run was not realistic. Deduplication by recording_id was applied at this stage: MusicBrainz can list the same recording under multiple track-release rows, and deduplication shifted the view from track-release to recording, which is the right level for lyric analysis.

The main enrichment output contained 1,082,070 recording-level entries, of which 81,823 had retrieved lyrics. A backfill pass targeted pre-1990 candidates, since older songs are less consistently represented on contemporary lyric platforms. Most retry batches returned nothing – itself a methodologically useful finding. A diagnostic check on the first 2,000 rows showed a 66.2% Genius found rate, 63.0% lyrics found rate and an average match score of 0.4368.

The reduction from millions of candidates to 81,823 records with lyrics is expected. Most MusicBrainz records have no matching Genius page, and many that do are duplicates, instrumentals, alternate versions or items where the metadata does not match well enough. The bias this introduces is real: songs available on Genius tend to be more popular, more recent and from genres with strong online lyric communities. Older chanson, archival recordings, regional repertoire and commercially marginal music are underrepresented. Diachronic findings should be read as trends within a lyric-available corpus, not as a map of all French musical output.

2.5. Working dataset

After enrichment, one more curation step produced a standardised working dataset of 66,711 rows with six columns: title, tag, artist, year, lyrics, language. These map directly onto the analytical needs:

title and artist identify the song, year enables diachronic grouping, tag provides genre context, lyrics contains the text, language gives a basic consistency check. The gap between 81,823 enriched records and 66,711 working rows reflects a change in purpose – the larger pool maximised acquisition, the smaller retains only records compatible with the analytical schema. “Year” was bound to [1925, 2025]. All rows already fell within this range, so the filter was a confirmation rather than a cut. The decade distribution, however, is starkly uneven, as Table 3 shows.

Table 3 – Working dataset distribution by decade

Decade	Entries	Decade	Entries
1920s	58	1980s	2,317
1930s	30	1990s	5,013
1940s	77	2000s	13,206
1950s	594	2010s	31,616
1960s	1,740	2020s	10,031
1970s	2,029		

This concentration in recent decades is predictable but consequential: decade-level comparisons need to account for unequal sample sizes, and early-period claims cannot be made with the same confidence as 2010s claims. Figure 1 visualises the same distribution. Missing-data rates were low overall: language 3.09%, tag 2.22%, lyrics 2.22%, title 0.003%. Language metadata had the highest missingness, but since the corpus had already passed French-likelihood filtering, dropping all language-null rows would have removed valid French songs with incomplete annotation.

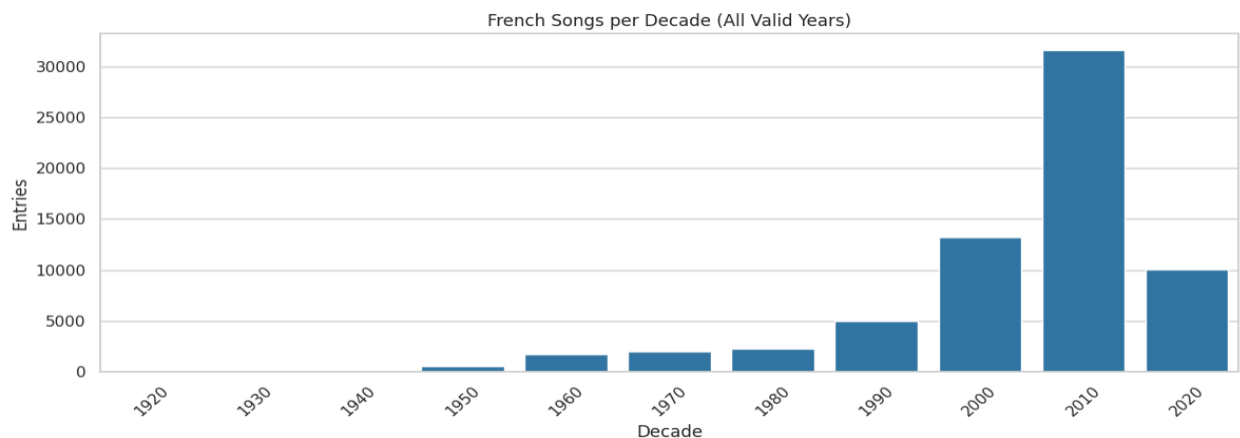


Figure 1 – French songs per decade in the working dataset (notebook 02, EDA)

2.6. Cleaning and normalization

Raw lyrics from web sources are noisy: section labels like [Chorus] or [Couplet], metadata lines (“Paroles de”, “Produced by”, “Contributors”, “Embed”, “You might also like”), embedded URLs, contributor counts, encoding artefacts. Left untreated, these flood topic models with structural vocabulary that has nothing to do with the song.

Cleaning kept three versions of each text: lyrics_raw (as loaded), lyrics_normalized (whitespace and URL handling) and lyrics_clean (section labels and metadata removed). Keeping all three makes the process auditable. Repeated lines and choruses were preserved – repetition is part of song lyrics, not noise – but the consequence for lexical frequency is handled later, in classification. The character-length statistics in Table 4 show that cleaning removed mainly small amounts of metadata rather than substantial lyrical content.

Table 4 – Character-length statistics before and after cleaning

Measure	Raw	Normalized	Cleaned
Mean	2,049.77	2,049.61	1,993.91
Median	1,747	1,747	1,698
25th percentile	1,058	1,058	1,040
75th percentile	2,760	2,759	2,669
Maximum	77,831	77,828	77,819

The difference between raw and cleaned mean lengths is about 56 characters, confirming that cleaning mostly stripped metadata. The very high maximum (~78,000 characters) flags entries with duplicated text or unusual structure that may need attention in robustness checks. After cleaning, 65,230 rows had usable text and 1,481 were excluded. The output was exported as nlp_final_dataset.csv and forms the input to the topic and mood analysis.

2.7. Exploratory analysis

Before any classifier ran, the cleaned corpus was examined on its own terms – composition, decade distribution, genre balance, lexical patterns. The temporal imbalance noted earlier is reinforced here: the 2010s alone hold 31,616 entries; the 1920s, 1930s and 1940s together hold 165. Early-decade results will be based on small samples and need cautious reading.

Genre composition is also unbalanced, as Table 5 shows. Rap and pop together account for the overwhelming majority of the corpus. This matters analytically. A rise in social critique, slang or lexical density across the century might reflect a real cultural shift – or it might track the growing share of rap. Trends that hold both across the whole corpus and within major genres are more convincing than ones that dissolve when genre is held fixed.

Table 5 – Genre composition (all decades)

Genre	Songs
rap	32,148
pop	28,665
rock	2,639
misc	2,400
rb	796
country	63

Genre composition by decade is shown in Figure 2; the rise of rap from absent before the 1990s to dominant in the 2010s and 2020s is the single largest compositional change in the corpus, and any decade-level trend has to be checked against it.

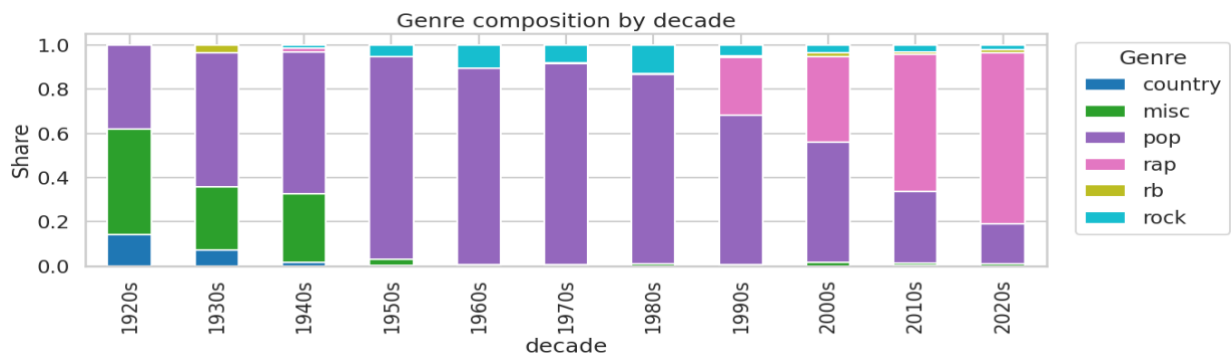


Figure 2 – Genre composition by decade (notebook 03, capped corpus)

Lexical richness – average unique words per song – was also computed by decade (Appendix A, Figure A1). The pattern is shaped by song length, genre and repetition, not by anything simple. It is shown for completeness but is not used as a finding about lyrical complexity.

2.8. Classification preparation

Lyrics differ from prose: choruses repeat, hooks return, ad-libs are formulaic. If a song with a chorus that fires five times is passed to a classifier raw, the chorus dominates the input and the verses get nothing. The classification notebook (03_topic_mood) therefore created an additional column, `lyrics_dedup`, in which repeated lyric lines were collapsed. The deduplication was conservative: lines under four characters (“non non non”, “ooh”) were not touched, because expressive fragments can carry meaning.

Tokenisation followed, with French stopwords and lyric fillers stripped. Songs with fewer than 40 valid tokens were removed. Duplicates by (artist, title, year, document text) were dropped, and decades labelled Unknown were excluded. This left 63,652 songs from the 65,230-song cleaned corpus – about 97.6%. Deduplication trimmed about 19% of raw text on average, bringing the mean classifier input to roughly 1,024 characters: enough to give the model two or three meaningful chunks per song.

Because transformers have hard input-length limits, classification used chunked inference rather than naïve truncation. Each lyric was split into overlapping 400-character windows with 50-character overlap, up to six chunks per song. Chunk-level scores were averaged into a single song-level distribution. Truncation would have been systematically biased toward openings; averaging keeps the rest of the song in the picture.

2.9. Topic and mood taxonomy

The taxonomy is fixed: eight topic labels, eight mood labels, all in French. The labels are in French because the classifier compares lyrics to natural-language hypotheses, and same-language hypotheses are semantically more natural for French texts. A fixed taxonomy across the century is what makes diachronic comparison possible – if labels reshuffled every decade, nothing would compare.

The topic labels and what they cover are shown in Table 6; the mood labels in Table 7.

Table 6 – Topic taxonomy

Topic label	Analytical meaning
amour et désir	Love, desire, romance, intimacy, attraction
rupture, perte et deuil	Breakup, separation, loss, grief, mourning
fête, plaisir et célébration	Partying, pleasure, celebration, joy of life
nostalgie, mémoire et temps qui passe	Memory, nostalgia, aging, time passing
révolte, critique sociale et engagement	Protest, social critique, politics, engagement
introspection, quête de sens et spiritualité	Self-reflection, existential questioning, spirituality
voyage, lieux et appartenance	Places, movement, migration, home, belonging
vie quotidienne, famille et société	Everyday life, family, ordinary social relations

Table 7 – Mood taxonomy and valence mapping

Mood label	Analytical meaning	Valence
joyeux	Joyful, cheerful, festive, optimistic	Positive
tendre	Tender, affectionate, intimate	Positive
apaisé ou plein d'espoir	Peaceful, hopeful, calm	Positive
nostalgique	Nostalgic, retrospective	Positive*
triste	Sad, sorrowful, melancholic	Negative
colérique ou rebelle	Angry, rebellious, confrontational	Negative
angoissé	Anxious, worried, fearful, tense	Negative
ironique ou désabusé	Ironic, disillusioned, cynical	Ambivalent

Valence is an operational grouping, not a philosophical claim. Nostalgia is placed in the positive group only for coarse cross-model agreement testing; in interpretation, it is treated as mixed affect. The hierarchy is used as a fallback when two models disagree on a fine-grained mood but still agree on direction.

Labels are connected to the classifier through hypothesis templates. The primary topic template is *Cette chanson évoque des thèmes liés à {}*; the primary mood template is *On peut percevoir dans cette chanson une tonalité {}*. The classifier then estimates how strongly each lyric supports each filled-in hypothesis. Two paraphrased templates per task are also kept for template-sensitivity testing in Section 2.11.

A fixed taxonomy simplifies. Songs can carry several topics at once and many moods overlap – protest can be introspective; love can be loss. The labels are dominant classifications, not exhaustive descriptions. A multi-label fallback (threshold 0.40) keeps additional labels above that score, so dominant-and-secondary tags are both available.

2.10. Zero-shot NLI framework

There is no labelled training set of French song lyrics for these eight topics and eight moods. Building one at the scale this corpus needs would take thousands of annotations across decades, genres and registers. Zero-shot natural-language inference [24] is the practical alternative: each lyric is treated as a premise, each label is inserted into a French hypothesis template, and the model estimates how strongly the premise supports each hypothesis.

Two models were used. The primary classifier (M1) is MoritzLaurer/mDeBERTa-v3-base-mnli-xnli – multilingual, trained for NLI, robust in French. The validation classifier (M2) is joeddav/xlm-roberta-large-xnli, run independently rather than as part of an ensemble. The point of running two is to expose where they agree and where they don’t. Agreement is evidence; disagreement is a useful signal about which labels are stable. Their outputs are also averaged into an ensemble after validation. Table 8 summarises the three layers.

Table 8 – Classification models and their roles

Model	Role
MoritzLaurer/mDeBERTa-v3-base-mnli-xnli	Primary classifier (M1)
joeddav/xlm-roberta-large-xnli	Validation classifier (M2)
Average of M1 and M2 score matrices	Ensemble used for final labels

Each song produced four kinds of output: a primary M1 prediction, an M2 prediction, an ensemble prediction, and a multi-label set above 0.40. A confidence policy – threshold 0.30 and a margin

threshold of 0.06 between the top and second labels – flagged weak or ambiguous predictions as uncertain rather than forcing a label. The classification parameters used throughout are listed in Table 9.

Table 9 – Classification parameters

Parameter	Value
Classified corpus size	63,652 songs
Text column for classification	lyrics_dedup
Chunk size	400 characters
Chunk overlap	50 characters
Maximum chunks per song	6
Confidence threshold	0.30
Margin threshold	0.06
Multi-label threshold	0.40
Topic labels	8
Mood labels	8

Zero-shot has clear advantages here: it applies the same labels consistently across the entire corpus, it does not need annotated training data, and it works from semantic label descriptions rather than word counts. It also has clear failure modes: it is sensitive to template wording, it confuses overlapping categories (nostalgia vs sadness; rebellion vs protest), and it doesn’t read irony, slang or genre conventions the way a human does. The validation framework in Section 2.11 maps where those failures are concentrated.

2.11. Validation and uncertainty

Four validation layers tested the outputs. None is decisive on its own; together they describe which labels deserve trust.

2.11.1. Template sensitivity

NLI models are template-sensitive. “This song evokes themes related to love” and “The main theme of this song is love” point at the same concept but can yield different scores. The primary model was run with three paraphrased templates per task, and agreement was measured with top-1 label

agreement and Cohen's κ . Across all three templates, agreement reached 35.6% for topics and 35.3% for moods. Pairwise κ ranged 0.32–0.54 for topics and 0.26–0.46 for moods. The lesson: fine-grained song-level predictions should not be overinterpreted, but decade-level aggregates absorb a lot of this noise.

Stability varied by label. Topic-wise, voyage and révolte were more stable than rupture and introspection. Mood-wise, joyeux and angoissé were most stable; apaisé ou plein d'espoir, ironique ou désabusé and colérique ou rebelle were essentially unstable.

2.11.2. Cross-model validation

M1 and M2 were compared on the full 63,652-song corpus using three metrics. Strict top-1 κ was 0.283 for topics and 0.162 for moods. Raw top-1 agreement was 39.0% (topics) and 34.2% (moods). Top-1 is a harsh measure here, because lyrics often map plausibly to several labels. Softer comparisons on the full score vectors gave a median Spearman ρ of 0.619 (topics) and 0.476 (moods) – the models often ranked candidate labels similarly even when they disagreed on the winner. Cross-model results are summarised in Table 10.

Table 10 – Cross-model agreement metrics (M1 vs M2)

Task	Top-1 κ	Top-1 agreement	Median Spearman ρ	Above $\rho = 0.5$
Topic	0.283	39.0%	0.619	64.3%
Mood	0.162	34.2%	0.476	47.3%

At the coarser valence level (positive / negative / ambivalent), mood agreement rose to 68.6% and $\kappa = 0.307$. Models often agreed on direction even when they disagreed on the precise mood. The per-label cross-model confirmation in Table 11 separates the taxonomy into reliable and unstable.

Table 11 – Per-label cross-model confirmation rates

Topic	Conf.	Mood	Conf.
rupture, perte et deuil	80%	nostalgique	74%
amour et désir	50%	joyeux	54%
nostalgie, mémoire et temps qui passe	45%	angoissé	52%
fête, plaisir et célébration	40%	triste	45%
introspection, quête de sens et spiritualité	37%	tendre	42%
voyage, lieux et appartenance	35%	apaisé ou plein d’espoir	5.6%
révolte, critique sociale et engagement	16%	ironique ou désabusé	1.6%
vie quotidienne, famille et société	6%	colérique ou rebelle	1.5%

Six topic labels and five mood labels passed practical reliability thresholds ($\geq 35\%$ for topics, $\geq 42\%$ for moods). The other three mood labels – anger/rebellion, irony/disillusionment, peacefulness/hope – performed badly. This is expected in lyrics, where these registers depend on tone, irony and performance more than on lexical surface.

2.11.3. Human annotation

A stratified subset of 182 songs was annotated by hand and compared with the ensemble. Topic classification reached $\kappa = 0.36$ and macro-F1 = 0.40. Mood classification was weaker, at $\kappa = 0.18$ and macro-F1 = 0.20, pulled down by exactly the three problematic mood labels above. When the mood evaluation was restricted to the five validated mood labels (angoissé, joyeux, nostalgique, tendre, triste), agreement jumped to $\kappa = 0.52$ and macro-F1 = 0.61. The same labels fail under template variation, cross-model agreement and human comparison – three independent checks converging on the same broken categories.

2.11.4. Confidence calibration

Calibration asks whether a model that reports 40% confidence is right about 40% of the time. Expected calibration error was 0.082 for topics – well-calibrated – and 0.130 for moods. Mood confidence becomes overconfident at the upper end, again largely because of the three broken labels. The

implication is straightforward: rank predictions by validated label, then by confidence, before claiming anything historical.

Validation together points to the same conclusion. The pipeline is useful but noisy. The noise is not random – it lives in a few specific labels. Aggregate, decade-level trends on the validated labels are the interpretable signal; everything else gets treated more cautiously.

2.12. Aggregation and statistical testing

Decade is the unit of interpretation, not the individual song. For each decade, the share of songs assigned to each topic and mood was computed. Comparing those shares across decades is what the trend analysis tests. Several controls were applied to make this honest.

Minimum decade size: 20 songs. Below this, the share is too unstable to test. Artist-decade cap: 30 songs per artist per decade – a small number of very productive artists can otherwise dominate a decade. Artist-level cluster bootstrap: songs by the same artist are not independent, so the bootstrap resamples artists, not songs, recomputing the shares across 1,000 iterations and taking the 2.5th and 97.5th percentiles as a confidence interval. The control parameters are summarised in Table 12.

Table 12 – Aggregation controls

Control	Value	Purpose
Minimum decade size	20 songs	Avoid unstable decade estimates
Artist-decade cap	30 songs	Reduce dominance of prolific artists
Bootstrap unit	Artist	Account for artist clustering
Bootstrap iterations	1,000	Estimate uncertainty intervals

Trend tests use the Mann–Kendall non-parametric test [25, 26] and Sen’s slope [28]. The series are short – eleven decade points – so a linear regression would be the wrong tool: too many assumptions for too little data. Mann–Kendall just asks whether the series goes up or down monotonically; Sen’s slope estimates by how much. Sixteen series were tested (eight topics, eight moods), so p-values were corrected with the Benjamini–Hochberg false discovery rate procedure [27].

The final robustness step is genre-adjusted decomposition. The corpus changes genre composition strongly across decades – rap goes from absent to dominant. If a topic looks like it’s rising because rap is rising, that’s a genre-mix effect, not a cultural trend. For each topic, two logistic

regressions estimate the decade effect: one with decade alone, one with decade plus genre. The interpretation rule is simple, as shown in Table 13.

Table 13 – Genre-adjustment interpretation rules

Pattern after genre control	Interpretation
Decade coefficient roughly unchanged	Trend is genre-robust
Decade coefficient much smaller	Trend is partly explained by genre shift
Decade coefficient near zero	Trend is mostly a genre-composition artefact
Decade coefficient reverses	Unadjusted trend was misleading
Model fails (singular matrix)	Label too sparse for genre adjustment; interpret cautiously

Combining FDR-corrected trend tests, cross-model validation and genre adjustment produces a three-tier hierarchy: robust trends (worth interpreting historically), partly confounded trends (worth reporting with the genre caveat) and unreliable trends (not main findings). Chapter 3 uses exactly this hierarchy.

2.13. Methodological limitations

The pipeline was designed to be transparent and cautious. Several limitations remain; none invalidates the analysis, but each defines where the results stop.

Corpus availability bias: the corpus represents songs whose lyrics are on Genius, which favours popular, recent, and digitally visible repertoire. Older chanson, obscure recordings and regional archives are underrepresented.

Historical imbalance: the 2010s contain roughly half the corpus; the 1920s–1940s contain a few dozen songs each. Early decades cannot be discussed with the same evidential strength as recent ones. The minimum-size threshold and bootstrap intervals reduce, but do not eliminate, this asymmetry.

Genre imbalance: rap and pop dominate; chanson, rock and country are minority categories. The genre-adjusted analysis exposes how much of an apparent trend is composition versus content. It exposes; it does not fix.

Metadata uncertainty: MusicBrainz is community-maintained. Year, country, language and genre fields are uneven. The composite French-likelihood score and deduplication reduce the impact, but no large metadata source is perfectly clean.

Matching uncertainty: title-and-artist similarity is approximate. Some matches are wrong, some are missed. The match-score weighting and threshold reduce this risk but do not eliminate it.

Classification uncertainty: zero-shot is template-sensitive and model-sensitive. Some labels – anger/rebellion, irony, peacefulness/hope, everyday life – are unstable across all three diagnostics. The thesis treats them differently from validated labels.

Emotion interpretation: a song classified as anxious is a song whose lyrics carry textual signals associated with anxiety. It is not a measurement of the artist’s state, the audience’s state, or French society’s state. This distinction is constitutive of the analysis and is preserved through every interpretation in Chapter 3.

Figurative language: lyrics traffic in metaphor, irony, slang, code-switching and performance. Automated classifiers do not handle these well. Reading still helps.

Language variation: 1920s chanson, 1960s yéyé, 2010s rap and 2020s multilingual urban music use very different French. Multilingual models help, but performance varies across registers, and this affects how comparable predictions are across decades.

Correlation vs causation: the analysis tests whether trends correspond to historical periods, not whether events caused lyrical changes. Chapter 3 uses cautious phrasing (“corresponds to”, “is consistent with”, “may reflect”) for exactly this reason.

The thesis does not claim to write a complete cultural history of French music. It builds a large, computationally structured analysis of available French song lyrics, with explicit controls for the most important sources of bias. The contribution is not eliminating uncertainty but making it visible.

2.14. Chapter conclusions

This chapter built the methodological base of the thesis by assembling a large-scale diachronic corpus of French song lyrics for the period 1925–2025. No suitable corpus of this kind existed, so the dataset had to be constructed from scratch through a multi-stage pipeline: MusicBrainz metadata, Genius lyric retrieval, filtering, cleaning, and NLP-oriented preparation. The classification-ready result contains 63,652 songs and is, on the available evidence, one of the largest computationally structured collections of French-language lyrics currently available for research.

A second argument ran alongside the construction work. Large-scale cultural analysis cannot proceed on the assumption that the data are clean or representative, and the thesis insisted on handling uncertainty openly rather than smoothing it away. The biases that surfaced during the pipeline are familiar but worth naming: unequal historical coverage, genre imbalance, gaps in metadata, and the obvious limits of what online lyric platforms actually contain. These do not invalidate the analysis. They mark the territory inside which its claims can be made.

For classification, the thesis used a zero-shot natural language inference framework and validated it through cross-model agreement, template-sensitivity testing, human annotation, confidence calibration, and genre-adjusted statistical controls. Some labels turned out to be stable and interpretable; others remained unreliable because the figurative, ambiguous quality of song lyrics resists categorical decisions. That split then governed the interpretive strategy of Chapter 3, where only validated and genre-robust trends are treated as substantive findings.

The broader claim is that computational NLP can be applied meaningfully to French song lyrics, provided it is paired with transparent preprocessing, explicit uncertainty estimation, and historically informed reading. The corpus, classification pipeline, and validation framework built here are the empirical and methodological basis for the diachronic analysis that follows.

Chapter 3. Results and Interpretation

3.1. Overview

Chapter 2 set up the framework. This chapter applies it to 63,652 French-language songs spanning 1925–2025 and asks which of the model’s diachronic patterns are robust enough to support historical interpretation. The unit of analysis is the decade. The interpretive logic is conservative: a trend is reported as a main finding only if it survives FDR-corrected trend tests, cross-model validation and genre adjustment together. Trends that pass only the statistical test get a caveat; trends that fail validation are exploratory at most.

Two short summaries before the details. On the topic side, the strongest finding is the rise of nostalgia and memory. On the mood side, anxiety rises sharply and tenderness, sadness and joy all decline. The popular framing – “lyrics get more negative” – is too crude. What actually happens is more specific: the dominant form of negative or mixed affect shifts from sorrow toward unease, and from intimate warmth toward retrospection.

3.2. Topic distribution by decade

The topic distribution shows continuity and rebalancing rather than rupture. *Amour et désir* is the most persistent label across the century – unsurprising for popular song. But its share is not stable: it dominates the earlier and mid-century decades and loses ground in later decades, as the genre mix opens up and rap and contemporary pop bring in stronger registers of memory, identity, place and social critique. Love does not disappear; it stops being the default centre of gravity.

The clearest rising trend on the topic side is *nostalgie, mémoire et temps qui passe*. Songs in this category use memory, retrospection, time passing, return and loss. Crucially, this is also the topic label that holds up best in validation and survives genre adjustment (Section 3.5). It is the most defensible thematic finding in the analysis.

Voyage, lieux et appartenance recurs visibly across decades but without a clear monotonic shape – a stable thematic domain rather than a rising or falling one. *Rupture, perte et deuil* is similar: consistently present, but not reorganising the corpus. *Fête, plaisir et célébration* appears throughout but declines slightly; the decline is later shown to be partly genre-driven.

Two labels need care. *Révolte, critique sociale et engagement* looks like it rises in later decades, which would fit a tidy story about politically engaged rap. The cross-model confirmation rate is 16% and the genre-adjusted analysis (Section 3.5) actually reverses the trend. *Introspection, quête de sens et*

spiritualité and vie quotidienne, famille et société are too weakly validated to be central findings. Both have cross-model rates below 40%; vie quotidienne sits at 6%.

Figure 3 shows the decade-by-topic share heatmap on the validated, capped corpus. Table 14 summarises the descriptive patterns.

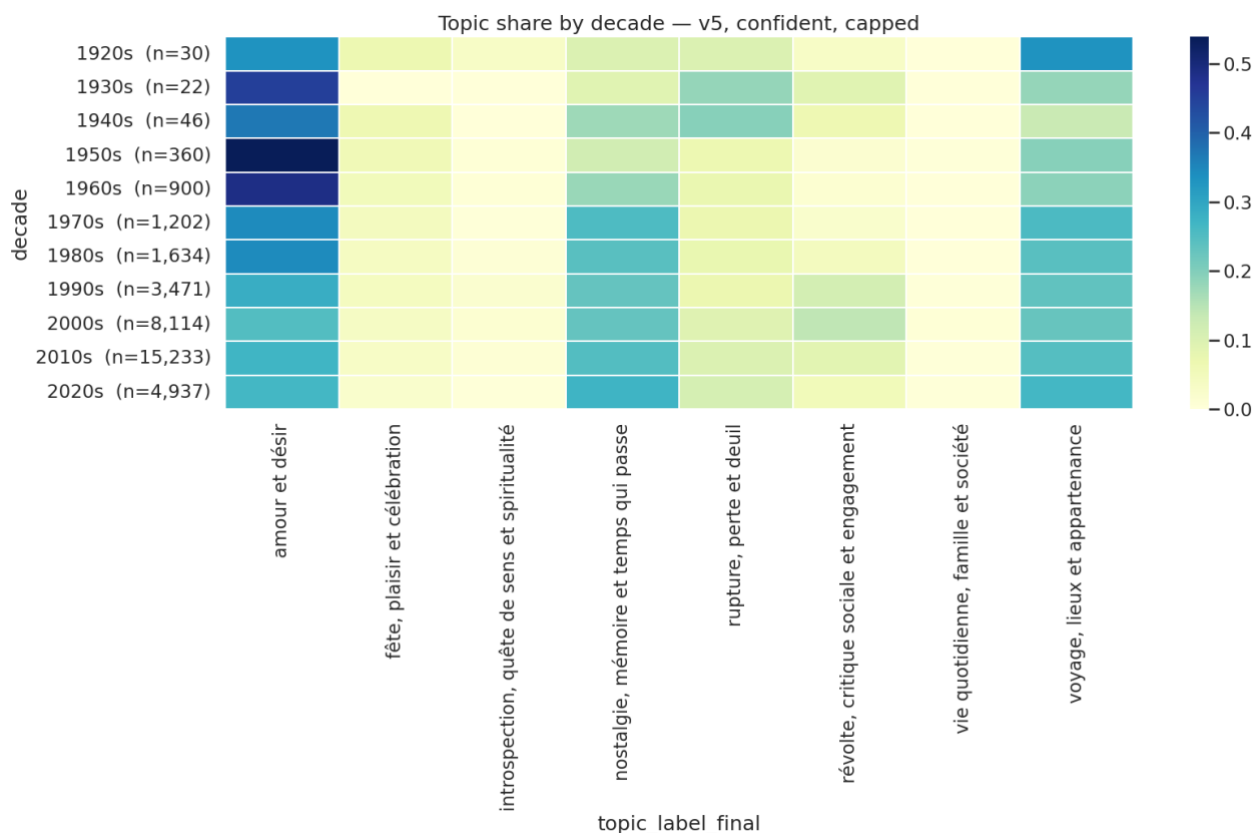


Figure 3 – Topic share by decade (capped corpus, validated labels)

Table 14 – Topic patterns across decades – descriptive summary

Topic	Pattern	Interpretive status
amour et désir	Dominant; relatively declining	Important, partly genre-confounded
nostalgie, mémoire et temps qui passe	Clearly rising in later decades	Strong thematic finding
voyage, lieux et appartenance	Recurring across decades	Stable, no clear monotonic trend
rupture, perte et deuil	Stable presence	Background theme

fête, plaisir et célébration	Slightly declining	Partly genre-confounded
révolte, critique sociale et engagement	Appears to rise	Reverses under genre control
introspection, quête de sens et spiritualité	Low, unstable	Exploratory only
vie quotidienne, famille et société	Very low, weak validation	Not a main finding

3.3. Mood distribution by decade

The mood side is where the strongest pattern lives. Five labels passed cross-model validation: *angoissé*, *triste*, *joyeux*, *nostalgique* and *tendre*. These are the ones to read.

The clearest descriptive result is the rise of *angoissé* as a dominant mood category. Anxiety-related tone takes a larger and larger share through the 2000s, 2010s and 2020s. This does not mean every recent French song is literally about anxiety. It means the classifier increasingly detects textual signals associated with tension, unease, worry, insecurity or emotional pressure. *Angoissé* performed well in validation, which gives this finding teeth: it is not just visible, it is reliable.

Anxiety is not the same as sadness. While *angoissé* rises, *triste* declines – and this is the part of the result that complicates a simple “more negative” narrative. Sadness becomes less dominant as the primary emotional label. The corpus does not become uniformly more negative; the form of negative affect shifts from sorrow to pressure.

Tendre – tenderness, softness, intimacy, affection – declines, and quite strongly. This is consistent with the relative decline of *amour et désir* on the topic side. *Joyeux* also declines, somewhat more weakly. The combination is striking: tenderness, joy and sadness all retreat, while anxiety expands. *Nostalgique* sits in the middle – emotionally mixed, persistent, in the analysis grouped operationally as positive but better read as ambivalent. Its persistence aligns with the rise of the nostalgia topic.

The remaining mood labels – *colérique ou rebelle*, *ironique ou désabusé*, *apaisé* ou *plein d’espoir* – are not reliable as standalone findings, for the validation reasons in Section 2.11. They appear in the visualisations for completeness, but the narrative around them in this thesis is deliberately thin.

Figure 4 shows the decade-by-mood heatmap. Table 15 summarises.

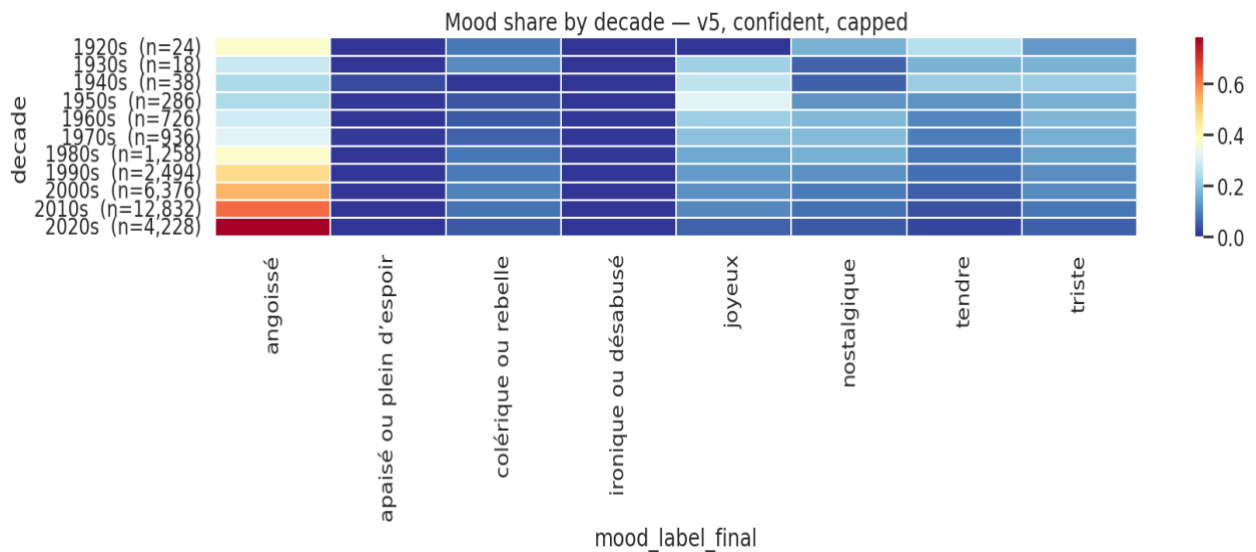


Figure 4 – Mood share by decade (capped corpus, validated labels)

Table 15 – Mood patterns across decades – descriptive summary

Mood	Pattern	Interpretive status
angoissé	Strongly rising	Central robust finding
tendre	Clearly declining	Central robust finding
triste	Declining	Reliable; partly migrates into anxiety/nostalgia
joyeux	Declining	Reliable, weaker than tendre/angoissé
nostalgique	Persistent, mixed affect	Read with the nostalgia topic
colérique ou rebelle	Low, unstable	Not reliable as a standalone finding
ironique ou désabusé	Low, unstable	Exploratory only
apaisé ou plein d'espoir	Low, unstable	Not suitable for strong claims

3.4. Statistically significant trends

Mann–Kendall and Sen’s slope, with BH-FDR correction across the sixteen series, identify eight significant trends at $q < 0.10$: four moods and four topics. The mood trends are stronger and more coherent than the topic trends. Tables 16 and 17 list them. Figure 5 shows the corresponding forest plot for moods.

Table 16 – Significant mood trends (Mann–Kendall + Sen’s slope, BH-FDR)

Mood	Direction	Sen’s slope (pp/decade)	Kendall τ	FDR q	Interpretation
tendre	Down	−1.64	−0.96	0.0008	Strong decline
angoissé	Up	+4.97	0.75	0.0148	Strong rise
triste	Down	−1.57	−0.67	0.0203	Clear decline
joyeux	Down	−2.22	−0.53	0.0585	Borderline decline

Tendre is the strongest single trend by τ – the most monotonic of the set. Angoissé has the largest absolute slope at +4.97 pp per decade. Triste declines at a similar rate to tendre. Joyeux is borderline but directionally clear. Together: the emotional centre of the corpus moves away from tenderness and direct sadness, toward anxiety, with joy also receding.

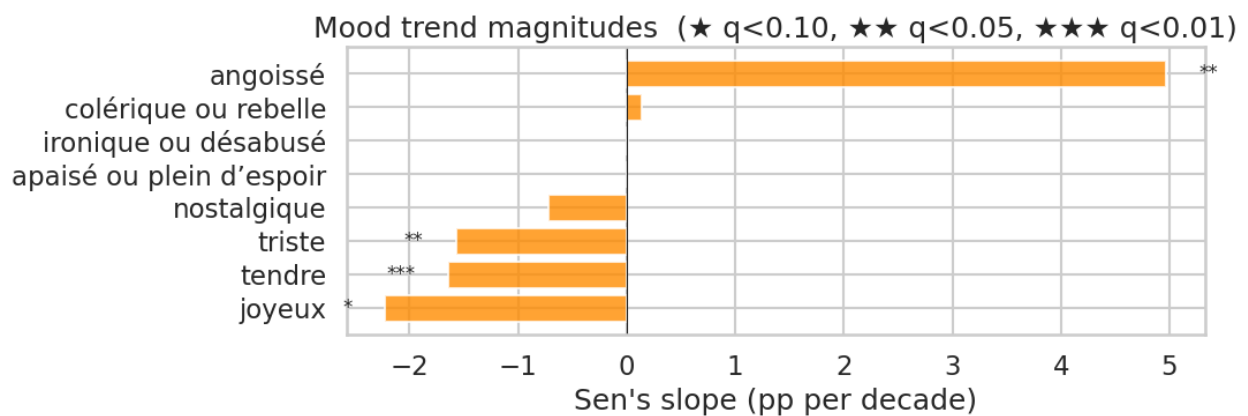


Figure 5 – Mood trend magnitudes (Sen’s slope; * $q < 0.10$, ** $q < 0.05$, * $q < 0.01$)**

Table 17 – Significant topic trends (Mann–Kendall + Sen’s slope, BH-FDR)

Topic	Direction	Sen’s slope	Kendall τ	FDR q	Interpretation
nostalgie, mémoire et temps qui passe	Up	+1.73	0.71	0.0165	Robust rise
fête, plaisir et célébration	Down	−0.42	−0.64	0.0260	Small decline
vie quotidienne, famille et société	Up	+0.04	0.60	0.0280	Statistically significant but unreliable
amour et désir	Down	−1.97	−0.53	0.0585	Borderline decline

Nostalgia is the standout topic trend – strong slope, high τ , and (as Section 3.5 shows) it strengthens under genre control. Amour et désir declines, but at the borderline of significance. Fête declines small but cleanly. Vie quotidienne shows a tiny positive slope (+0.04 pp/decade) that passes FDR but has a cross-model confirmation of 6% and a singular-matrix failure in genre regression – significance alone is not enough, and this trend is reported but not interpreted.

Eight other labels – mood: nostalgique, ironique, colérique, apaisé; topic: voyage, révolte, introspection, rupture – did not show significant monotonic trends after correction. That is not the same as saying they are absent. Several fluctuate around historical events, are concentrated in specific genres, or vary too non-monotonically for Mann–Kendall to catch.

3.5. Genre-confound analysis

A trend test can only get you so far when the corpus changes its genre composition across decades. Logistic regression on each topic – first with decade alone, then with decade plus genre – exposes how much of an apparent trend is composition. Table 18 reports the results.

Table 18 – Decade effect on topics, unadjusted vs genre-adjusted

Topic	Unadjusted β	Adjusted β	Attenuation	Interpretation
révolte, critique sociale et engagement	0.088	−0.187	313% (reversal)	Reverses under genre control
amour et désir	−0.139	−0.041	70.5%	Strongly attenuated

fête, plaisir et célébration	−0.125	−0.042	66.4%	Strongly attenuated
rupture, perte et deuil	0.097	0.033	66.3%	Strongly attenuated
voyage, lieux et appartenance	0.054	0.030	44.2%	Partly attenuated
nostalgie, mémoire et temps qui passe	0.060	0.075	−24.4% (strengthens)	Genre-robust; the strongest topic finding
introspection, quête de sens et spiritualité	—	—	—	Model failed: singular matrix
vie quotidienne, famille et société	—	—	—	Model failed: singular matrix

The reversal of *révolte* is the headline result here. The unadjusted decade coefficient is positive: protest themes appear to rise across the century. After controlling for genre, the coefficient becomes negative. The apparent rise of protest is largely the rise of rap in the corpus. Within any single genre, *révolte* does not increase. This does not make protest unimportant – it is tightly bound to rap and to the cultural work rap has done in French popular music since the 1990s. But it means the corpus-wide finding has to be stated precisely: protest becomes more visible because the genre that foregrounds it becomes more common, not because all French songs became more political.

Amour and *fête* show large attenuations (70.5% and 66.4%). Both decline genuinely, but the magnitude of the decline shrinks a lot once genre is controlled – much of it is the displacement of pop ballads and *chanson* by rap and rock. Voyage attenuates less; rupture attenuates similarly to amour but did not show a significant Mann–Kendall trend in the first place. Introspection and *vie quotidienne* can't be tested at all because of singular-matrix errors – their distribution is too sparse or too perfectly separated across genre cells.

Nostalgia goes the opposite way: the coefficient strengthens under genre control. That makes it the most genre-robust topic in the analysis – the only one where the trend is not even partly explained by composition. It is the single most defensible diachronic claim the thesis can make.

3.6. Historical interpretation

Three robust transformations come out of the analysis. They are not a complete history of French *chanson* – but they are the parts the data actually supports.

Decline of intimate-positive lyrical centrality

Tendre, joyeux and amour all retreat. Love, tenderness and joy do not disappear from French lyrics – they remain large categories – but they stop being the dominant register. Across the early and mid-century, romantic attachment and intimate address organise much of the corpus. By the later decades, the lyrical voice more often locates itself in tension, place, memory, or social pressure. This shift coincides with the broader transformation of French popular music from chanson-centred intimacy toward a heterogeneous field shaped by pop, rock and (especially) rap. Reading this as the death of love in French music is too dramatic; the more accurate description is that love stops being the default.

Rise of anxious and retrospective affect

Angoissé rises by roughly five percentage points per decade. Nostalgia rises by 1.7 points and stays strong under genre control. The combination is not arbitrary: anxiety points toward uncertainty about the present; nostalgia points toward a heightened relation to the past. Together they describe a cultural register where the present feels pressured and the past is emotionally charged. The pattern fits the broad social context of the late twentieth and early twenty-first centuries – precarity, urban transformation, globalisation, media intensification, terrorism, ecological worry, pandemic – without proving direct causation. It is consistent with that context, and it is consistent across multiple validation diagnostics. That is as much as the data can carry, and it is enough.

The decline of triste reframes the result. If “negative emotion” were treated as a single block, sadness should rise alongside anxiety. It doesn’t. Sadness is a localised, object-oriented feeling – loss, mourning, a specific pain. Anxiety is more diffuse and forward-facing. The shift from triste to angoissé is not just “more darkness”; it is a change in how negativity is expressed. That distinction matters historically.

Genre-mediated expansion of social critique

Révolte rises in the unadjusted view and reverses in the adjusted one. The honest claim is therefore narrower: protest does not increase across French song as a whole. It becomes more visible in the corpus because rap – the genre most likely to foreground protest, urban experience, exclusion and identity – becomes more common. Hammou’s history of French rap [30] documents the cultural mechanism behind this: rap as a politicised lyrical space built in the banlieues from the late 1980s. The genre transformation is itself a historical fact; treating it as a uniform century-long shift across all French music is the mistake the genre-adjusted regression catches.

Two further notes. The early decades of the corpus (1920s, 1930s, 1940s) hold too few songs to support decade-specific claims about wartime or interwar lyrical change. The trends in this thesis are over the full series, not localised event windows; finer-grained event analysis would need additional historical data and is left for future work (Section 3.7). And the model’s difficulty with anger, irony and rebellion is not the same as those things being absent from French lyrics – they are demonstrably present, especially in rap, but they are not separable cleanly by zero-shot NLI. Reading helps where models can’t.

3.7. Hypothesis evaluation and prospects

The original hypothesis was that periods of social and political turmoil are associated with an increased frequency of negatively valenced emotions and more reflective or introspective thematic content. The evidence supports part of this and forces a reformulation of the rest.

The rise of *angoissé* supports the negative-affect part of the hypothesis, but only after specifying which negative affect: anxiety, not sadness. The rise of nostalgia and memory supports the reflective part. The decline of *triste* contradicts a simpler version of the hypothesis that treats negative emotion as a single block. The instability of the *colérique* and *ironique* labels prevents any strong claim about a general rise in anger or irony, even though both are obviously present in the corpus.

A defensible reformulation: across the available corpus, French song lyrics show a long-term shift toward anxiety and retrospection, while moving away from tenderness, joy and direct sadness as dominant affective registers. This shift is not a simple increase in negativity, and several apparent thematic changes – most notably the rise of protest – are mediated by changes in genre composition. The thesis claims this much, and no more.

Several directions extend the work. The earliest decades (1920s–1940s) are underrepresented because of digital availability; future iterations could pull from the BnF, INA and digitised songbooks. The decade-level frame could be replaced or supplemented by event windows around the Second World War, May 1968, the oil shocks, the rise of French rap in the 1990s, the 2005 banlieue riots, the 2015 attacks, COVID-19 – finer-grained tests of when the trends turn. Genre-specific modelling – separate analyses for *chanson*, pop, rock, rap – would let within-genre change be measured directly. A supervised classifier, trained on a manually annotated French-lyric benchmark, would address the labels that zero-shot handles worst. Lexical analysis (*verlan*, Anglicisms, pronoun use, rhyme structure) would expand the picture beyond topic and mood. And cross-language comparison with German [10], K-pop [7] and Mandarin [11, 12] would test whether the rise of anxiety and nostalgia is French-specific or part of a wider shift.

The full code and notebooks for the pipeline, the cleaned dataset and the trained classifier outputs are available at <https://github.com/daniilmv/hsemasters> [31]. Anyone who wants to extend, replicate or argue with the results can start there.

3.8. Special project: Chanson

In parallel with the thesis, the results are presented through a public-facing project at <https://daniilvoloshin.com/chanson>. The thesis carries the methodology, validation and limitations; the website turns the findings into a cultural narrative. The two are complementary: one for examiners and researchers, one for music enthusiasts and French learners.

The project sits at the crossroads of digital humanities, French music culture and language learning. It lets users explore a century of French lyrics through themes, moods and decades. Stakeholders fall into a few groups, summarised in Table 19.

Table 19 – Chanson project – stakeholder structure

Stakeholder group	Interest	Project value
French music enthusiasts	Patterns in French music history	Visual and narrative exploration of topics, moods, decades
French language learners	Vocabulary, idiom, emotional expression via songs	Topic-/mood-based discovery of culturally meaningful lyrics
Teachers and educators	Materials for culture-aware language teaching	Curated thematic access across periods
Digital humanities researchers	NLP applied to cultural texts	A reproducible computational lyric pipeline [29]
Musicologists and cultural historians	Long-term shifts in themes and affect	Empirical entry point for qualitative interpretation
Cultural institutions	Accessible heritage presentation	Data-driven storytelling around French music
Playlist curators / platforms	Mood- and theme-aware discovery	Historical or thematic French music journeys

General public	French culture without technical background	Accessible narrative interface
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The project keeps the cautious framing of the thesis. Methodological notes flag that the corpus is built from lyrics available online, that early decades are underrepresented, that genre composition affects every trend, and that emotion labels refer to textual signals rather than psychological states. Without that transparency, the visualisations would risk becoming confident-looking claims about something the data cannot establish. With it, the project works as an example of computational cultural analysis presented responsibly to non-specialists.

3.9. Chapter conclusions

Several long-term thematic and emotional changes in French song lyrics emerged from the analysis in this chapter. The strongest results are emotional: anxiety-related affect rose substantially across the later decades of the corpus, while tenderness, joy, and direct sadness declined as dominant registers. Thematically, the clearest and most genre-robust long-term trend is the rise of nostalgia, memory, and retrospection.

Corpus composition and genre change matter more than is usually acknowledged. Several apparent trends weakened or disappeared once genre was controlled for, and the most striking case is protest and social critique. What looked like a rise turned out to depend almost entirely on the growing share of rap in the later decades, rather than on a uniform change across French song as a whole. Statistical trend analysis and genre-aware reading have to be done together; running one without the other produces misleading results.

It would also be wrong to say that French song lyrics simply become "more negative" over time. The form of negative affect changes. Sadness declines while anxiety rises, which suggests a move away from localized, object-oriented emotion toward more diffuse tension and psychological pressure. Nostalgia grows alongside this shift, pointing to a stronger preoccupation with memory and retrospection in contemporary lyrical culture.

So the value of computational NLP for cultural and linguistic change is real, but partial. Automated classification can reveal large-scale patterns that close reading alone would miss, while interpretation still depends on validation, historical context, and caution. The contribution of the thesis is therefore as much an analytical framework as a set of trends: a way of working with poetic and culturally situated texts at scale that does not pretend the texts are easier than they are.

Conclusion

This thesis examined the thematic and emotional evolution of French song lyrics between 1925 and 2025 through computational NLP analysis combined with historically informed interpretation. The starting point was a specific gap in the literature. English-language popular music has been studied extensively with large-scale computational methods; French song lyrics have rarely been analysed within a reproducible diachronic framework spanning an entire century.

To address that gap, the study built a large-scale corpus of French-language lyrics from open digital sources. MusicBrainz was the metadata backbone; Genius supplied lyric enrichment through large-scale title-and-artist matching. The classification-ready corpus contained 63,652 songs across eleven decades. Its distribution is historically uneven and shaped by digital availability bias, but it still supports one of the broadest computational analyses of French song lyrics currently available.

The methodological contribution is a transparent analytical pipeline designed for poetic and culturally situated text. Because no labelled dataset for French song lyrics existed, the thesis used a zero-shot natural language inference framework with multilingual transformer models. Topic and mood classifications were validated through several independent procedures: template-sensitivity testing, cross-model agreement, human annotation, confidence calibration, and genre-adjusted statistical controls. Automated outputs were not taken as objective truth. They were treated as probabilistic cultural signals whose reliability had to be checked before any interpretation could rest on them.

The empirical analysis produced several robust long-term trends. The strongest emotional finding is the substantial rise of anxiety-related affect in later decades, accompanied by declines in tenderness, joy, and direct sadness as dominant registers. The clearest thematic trend is the rise of nostalgia, memory, and retrospection. Read together, the results suggest that contemporary French song lyrics increasingly articulate emotional uncertainty and retrospective reflection, where earlier periods were more often organised around intimate, romantic centrality.

Several apparent cultural trends turned out to be heavily mediated by changes in genre composition. The supposed rise of protest and social critique, for instance, largely disappeared once genre controls were introduced. It was driven mainly by the growing prominence of rap rather than by a uniform change across all French music. Within cultural corpora, genuine diachronic change has to be distinguished from compositional effects, and the difference is not always visible at first.

Q1 asked how dominant themes evolved. The answer is gradual reorientation rather than rupture. Love recedes as a structural default, memory and place gain ground, social critique becomes more visible

through the genre that carries it, not through a corpus-wide turn. Q2 asked how emotional tones correspond to historical transformations. The evidence does not support event-level claims: the corpus is too unevenly distributed across time, and the early decades are too small for that kind of test. What it supports is a structural correspondence. The rise of anxiety and retrospection across the late twentieth and early twenty-first centuries fits the social texture of those decades: economic precarity, globalisation, urban change, political disillusionment, ecological worry. The analysis uses "is consistent with" deliberately. It cannot say these events caused the shifts; it can say the shifts are not random, not explained by genre composition alone, and temporally aligned with the broader context.

The research hypothesis is therefore only partially confirmed. The results support the idea that periods of broader social and cultural instability correspond to increased anxiety and to more reflective, retrospective lyrical content. The corpus does not, however, show a simple linear increase in negativity. The dominant form of negative affect changes over time: sadness falls, anxiety rises. The picture is one of structural transformation in emotional expression, not of a uniform deterioration of lyrical sentiment.

The limitations are worth stating directly. The corpus depends heavily on the availability of lyrics on contemporary online platforms, which favours recent, popular, and digitally visible repertoire. Early decades are underrepresented. Genre imbalance, especially the dominance of rap and pop in the 2000s and 2020s, affects all aggregate trends. Automated classifiers still struggle with irony, metaphor, code-switching, and the genre-specific language practices typical of song lyrics. None of this overturns the findings, but it does constrain what can be claimed from them.

Within those constraints, computational NLP can contribute meaningfully to cultural history when it is paired with careful validation and contextual reading. The corpus, codebase, and pipeline developed in this project are reusable for future work in digital humanities, corpus linguistics, musicology, and computational cultural analysis.

Future research could extend the corpus through archival sources such as the Bibliothèque nationale de France or the Institut national de l'audiovisuel, which would substantially improve coverage of the early twentieth century. Other directions worth pursuing include supervised classifiers trained specifically on French lyrics, genre-specific diachronic analyses, finer-grained event-based temporal modelling, and comparative cross-cultural studies involving other musical traditions. The decade-level frame could be replaced by event windows: the Second World War, May 1968, the oil crises, the rise of French rap in the 1990s, the 2005 banlieue riots, the 2015 attacks, COVID-19. Each is a candidate for an inflection-point test, provided the corpus covers the surrounding years densely enough.

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Appendix A. Key figures

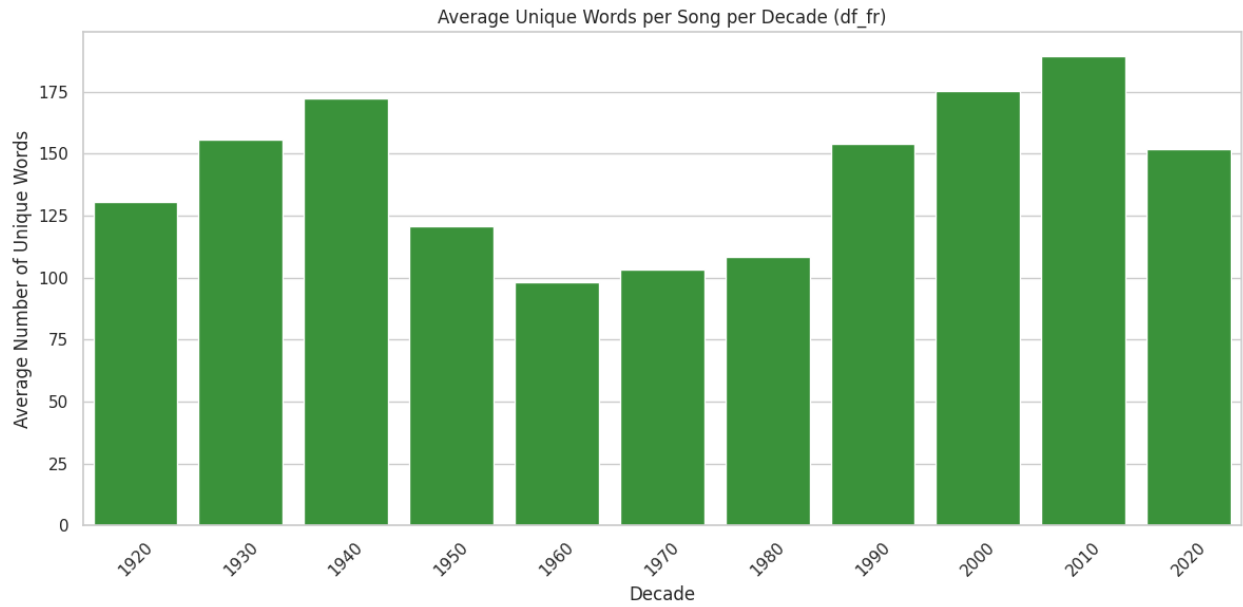


Figure A1 – Average unique words per song per decade (notebook 02)

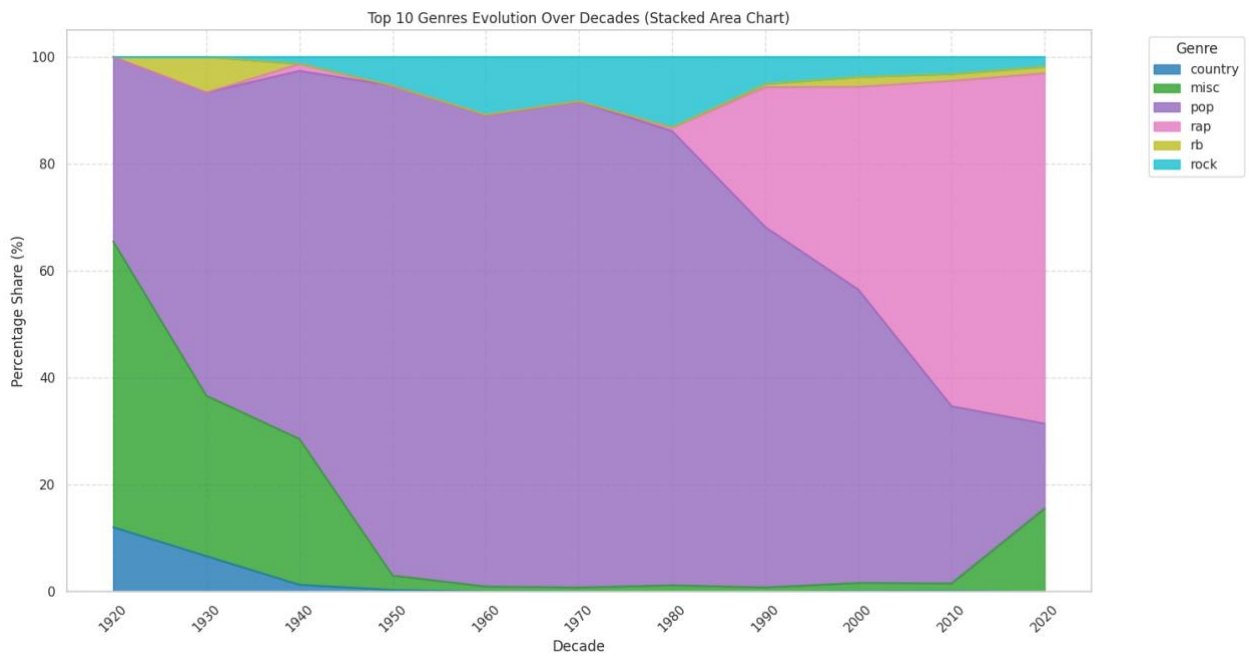


Figure A2 – Top 10 genres – share by decade (stacked area, notebook 02)

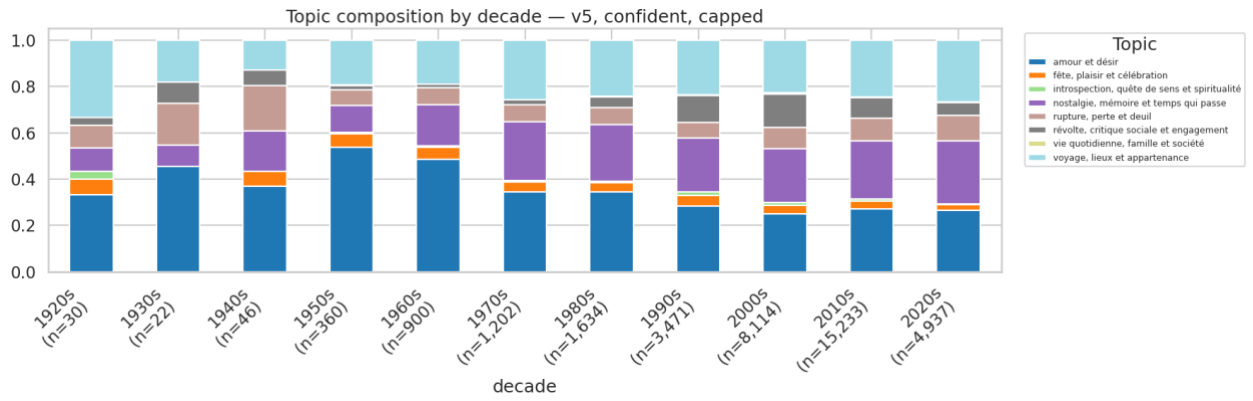


Figure A3 – Topic composition by decade (stacked bar, validated capped corpus)

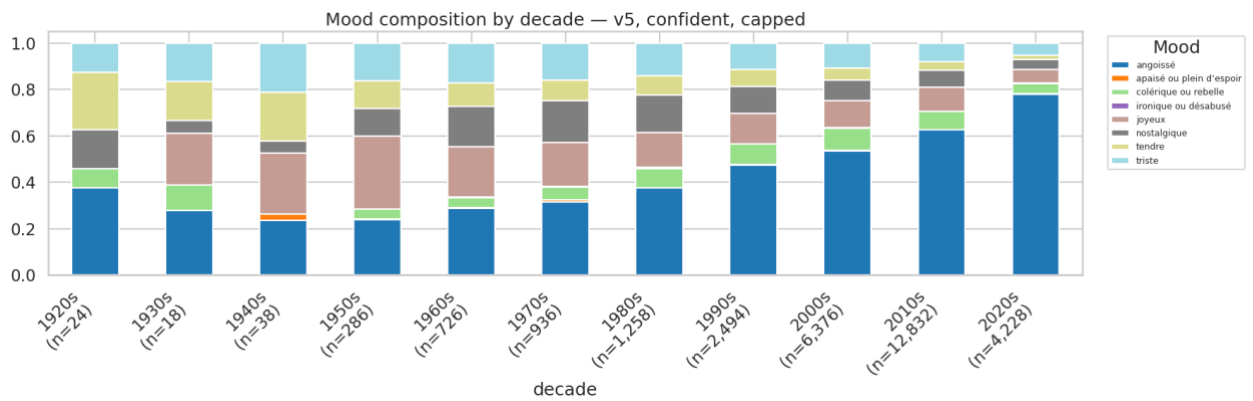


Figure A4 – Mood composition by decade (stacked bar, validated capped corpus)

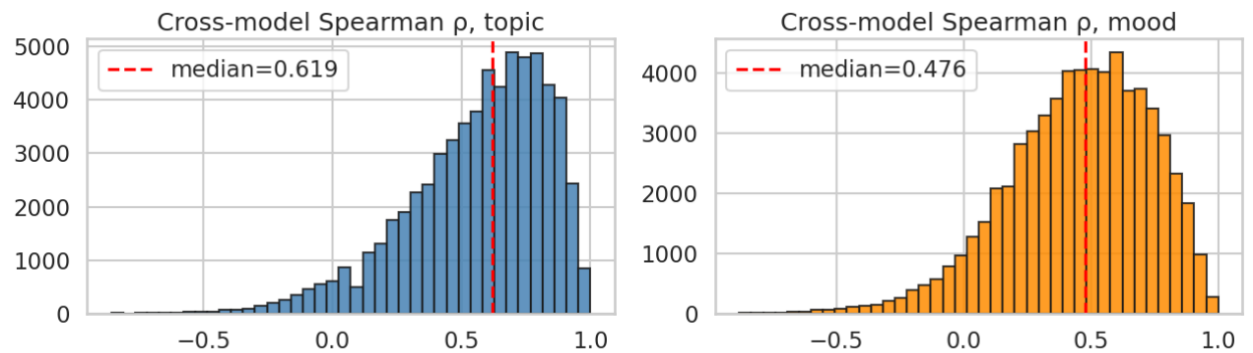


Figure A5 – M1 vs M2 cross-model Spearman correlation distributions

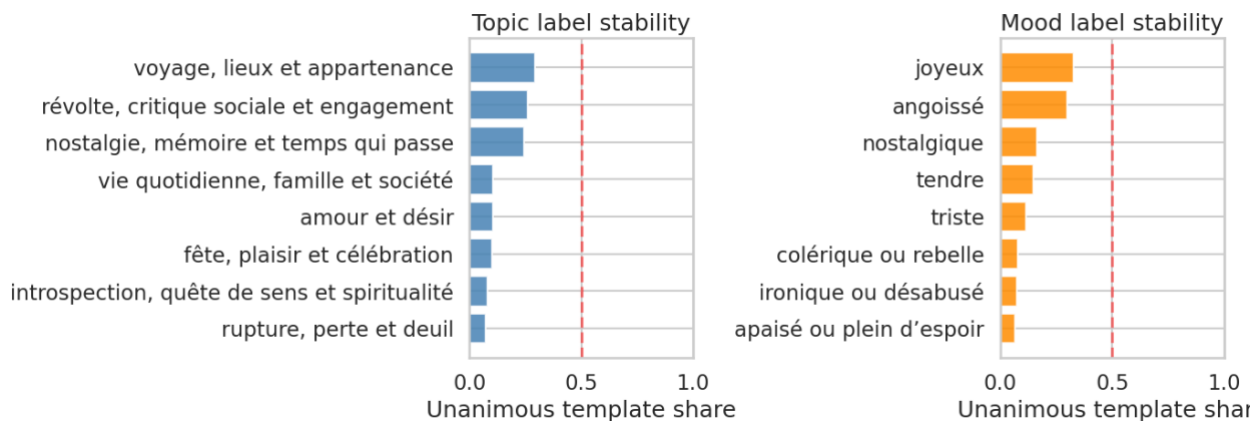


Figure A6 – Template stability – agreement under paraphrased hypothesis templates

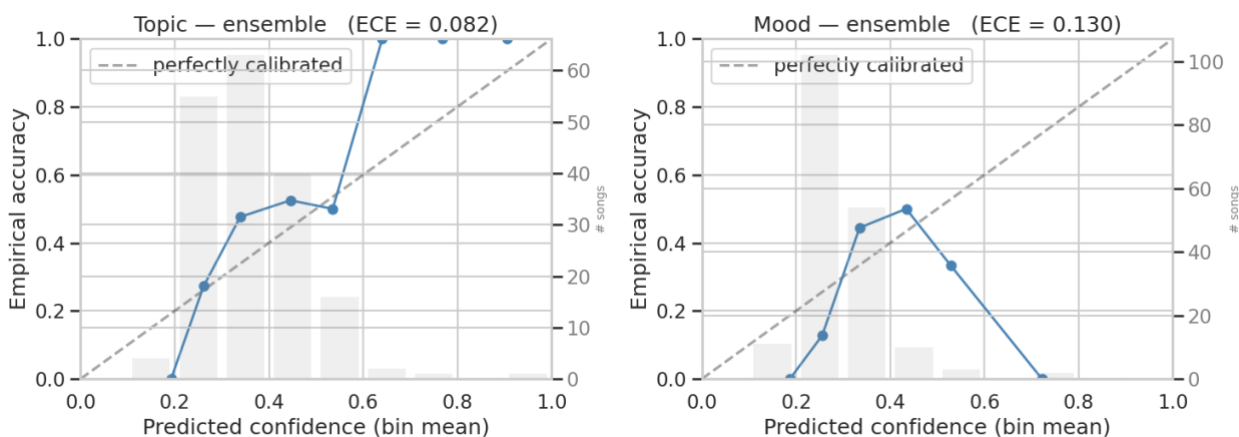


Figure A7 – Reliability diagrams (calibration) for topic and mood predictions

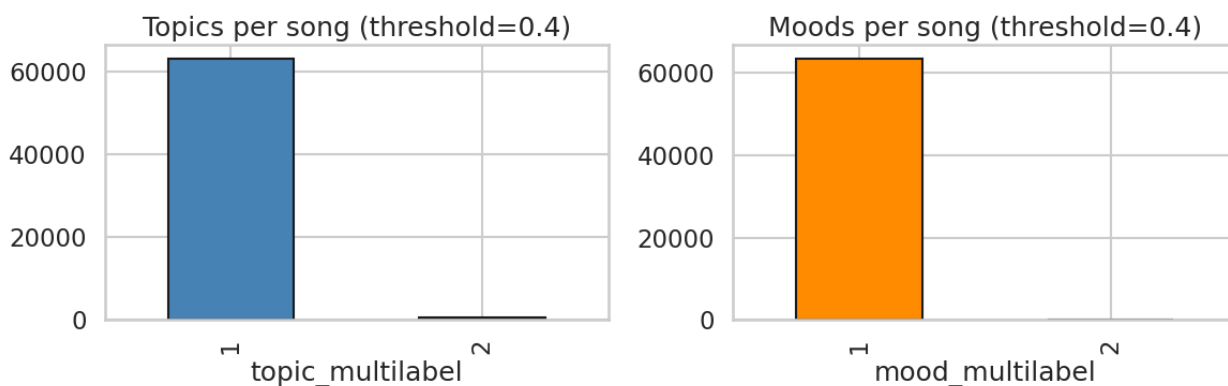


Figure A8 – Distribution of labels per song under the 0.40 multi-label threshold

Appendix B. Extended tables

Table B1 – Decade-level corpus shape vs classification-ready subset

Decade	Working dataset (n)	Classified n (capped)	Note
1920s	58	30	Small sample – interpret with caution
1930s	30	22	Small sample – interpret with caution
1940s	77	46	Small sample – interpret with caution
1950s	594	360	Postwar chanson period
1960s	1,740	900	Yéyé, chanson, early rock
1970s	2,029	1,202	Singer-songwriter era
1980s	2,317	1,634	Pop and variety dominance
1990s	5,013	3,471	Emergence of French rap
2000s	13,206	8,114	Rap and pop coexistence
2010s	31,616	15,233	Streaming era, rap surge
2020s	10,031	4,937	Post-pandemic, continued rap dominance

Capped n applies the 30-songs-per-artist-per-decade cap from Section 2.12; sums therefore fall below the 63,652 classification-ready total.

Table B2 – Convergence of three independent validation diagnostics on weak mood labels

Mood label	Template all-3 agree	M1↔M2 confirmation	Gold-standard F1
joyeux	Stable (33%)	54%	Solid
angoissé	Stable (30%)	52%	Solid
nostalgique	Solid	74%	Solid
triste	Solid	45%	Solid
tendre	Solid	42%	Solid
apaisé ou plein d'espoir	Unstable (7%)	5.6%	0.00
ironique ou désabusé	Unstable (8%)	1.6%	0.00
colérique ou rebelle	Unstable (8%)	1.5%	0.22

Three independent diagnostics flag the same three mood labels as broken. The convergence is what makes the reliability claim defensible.